

HPM5361-FOC

文件: HPM-FOC-G6618.kicad_sch

REV:0.1V

完全国产化选型

关键器件及参数:

MCU: HPM5361IEG1 RISC-V @ 480 MHz, 1714 CoreMark,
集成 FPU/DSP、16-bit ADC 与高性能运动控制外设。

三相驱动半桥 NMOS : CRSM038N10N4 100V 100A 3mΩRds(on)

栅极驱动器 : FD6288 250v耐受

辅助供电 : 5.5V~100V宽压buck, 栅极驱动供电由5V增压至12v, 宽工作范围

相电流检测: 三电阻低测采样 GAIN:7.5 Rshunt:2mΩ Range:~±100A
使用TPA6584Q-TS2R-S运放 Quad RRIO @ 10 MHz, 8 V/μs, ±0.3 mV typ Vos。

母线电压采样 + NTC 温度采集

一路FAULT输入, 由于硬件电路触发保护
一路GPIO, 用于控制LED指示状态

通讯接口:

- 1 FDCAN (主控制接口)
- 1 RS485 (可使用Tamagawa over SoC UART, 外接对应的编码器)
- 1 SPI (两个片选IO, 可复用为ABZ输入)
- 1 JTAG 调试接口
- 1 UART 上位机调参接口

REV:0.2V

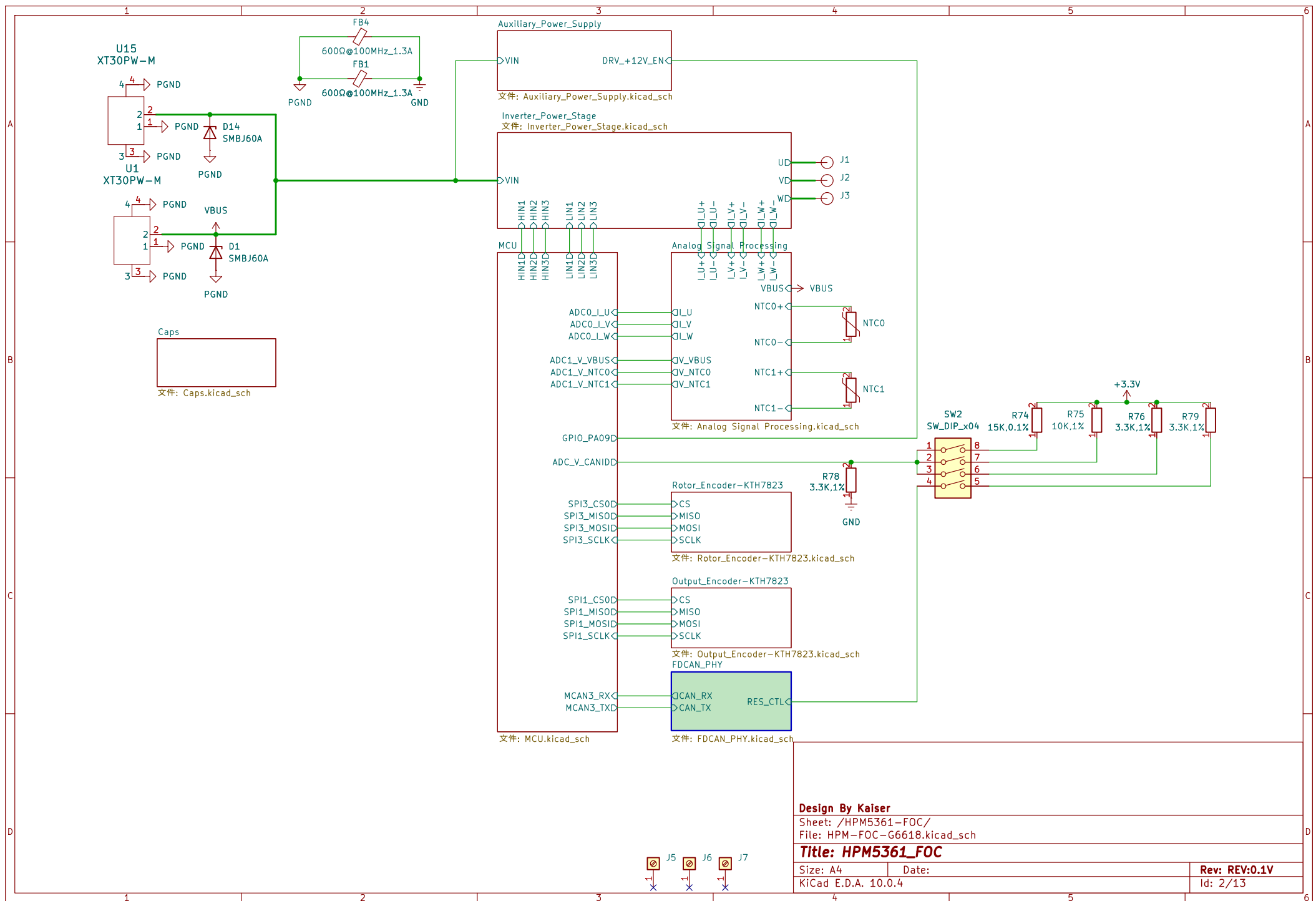
调整编码器接口:
UART硬件流控的高速RS485
QEI(U V W)模式, MCU硬件旋变解调

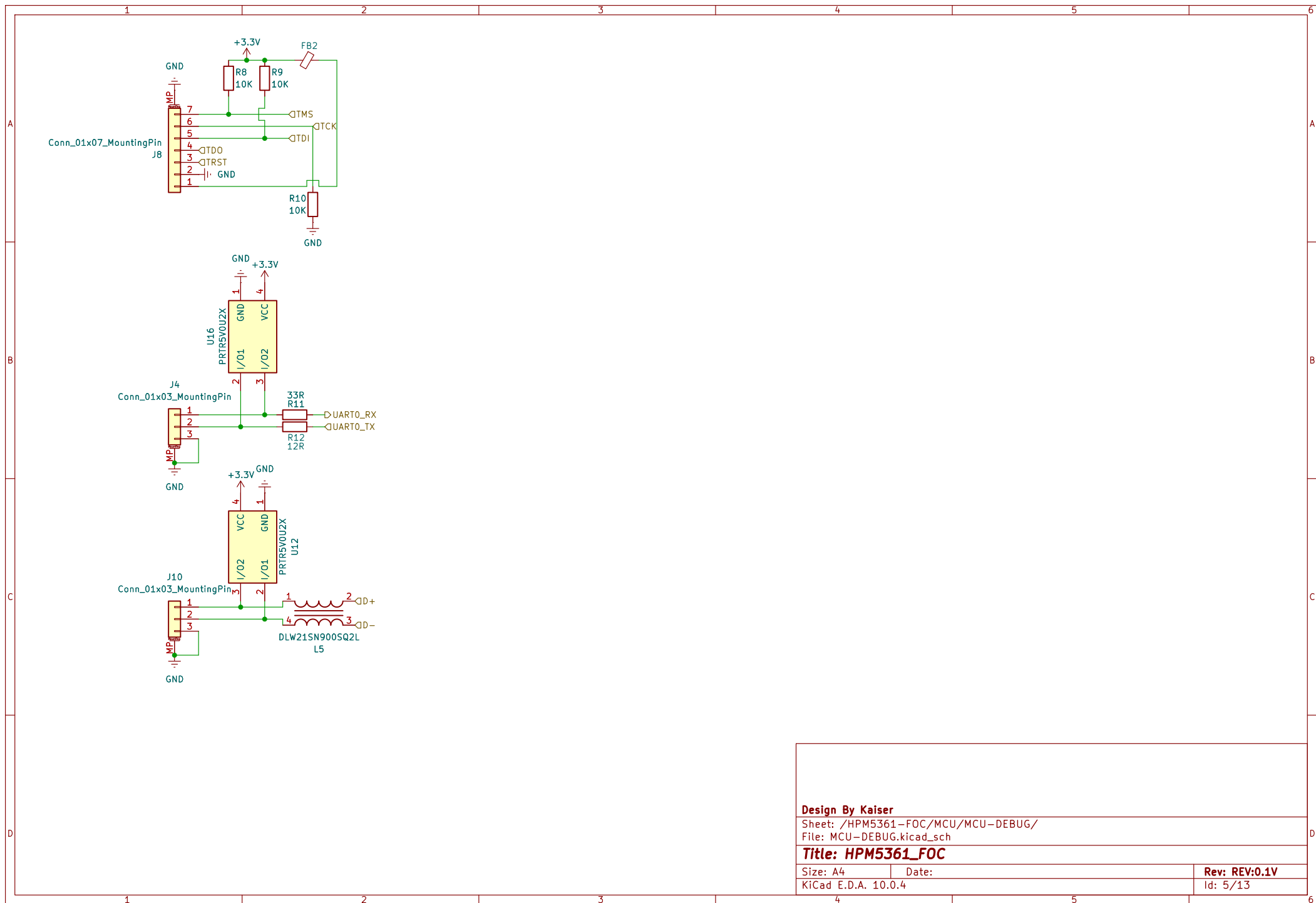
Design By Kaiser

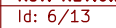
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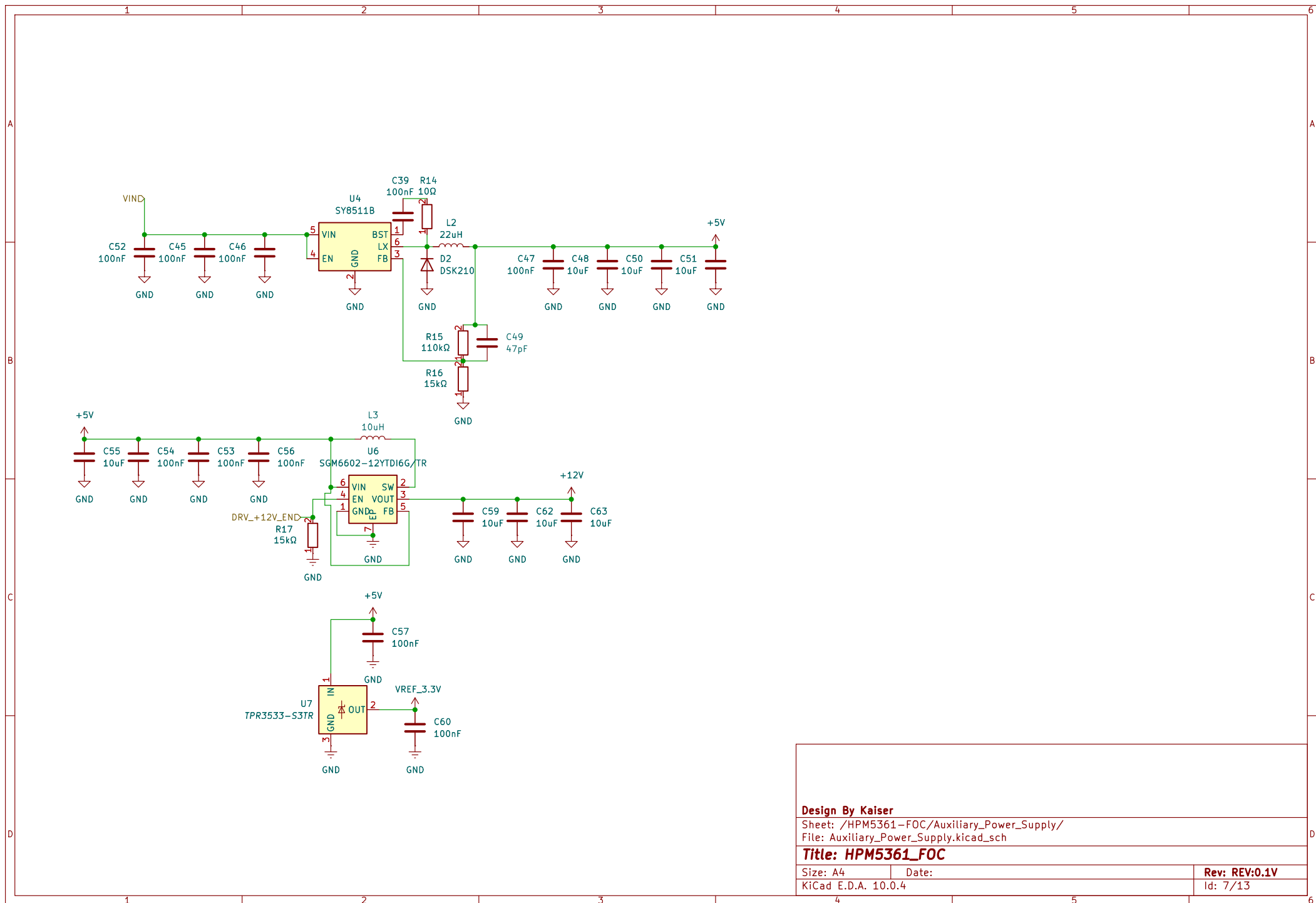
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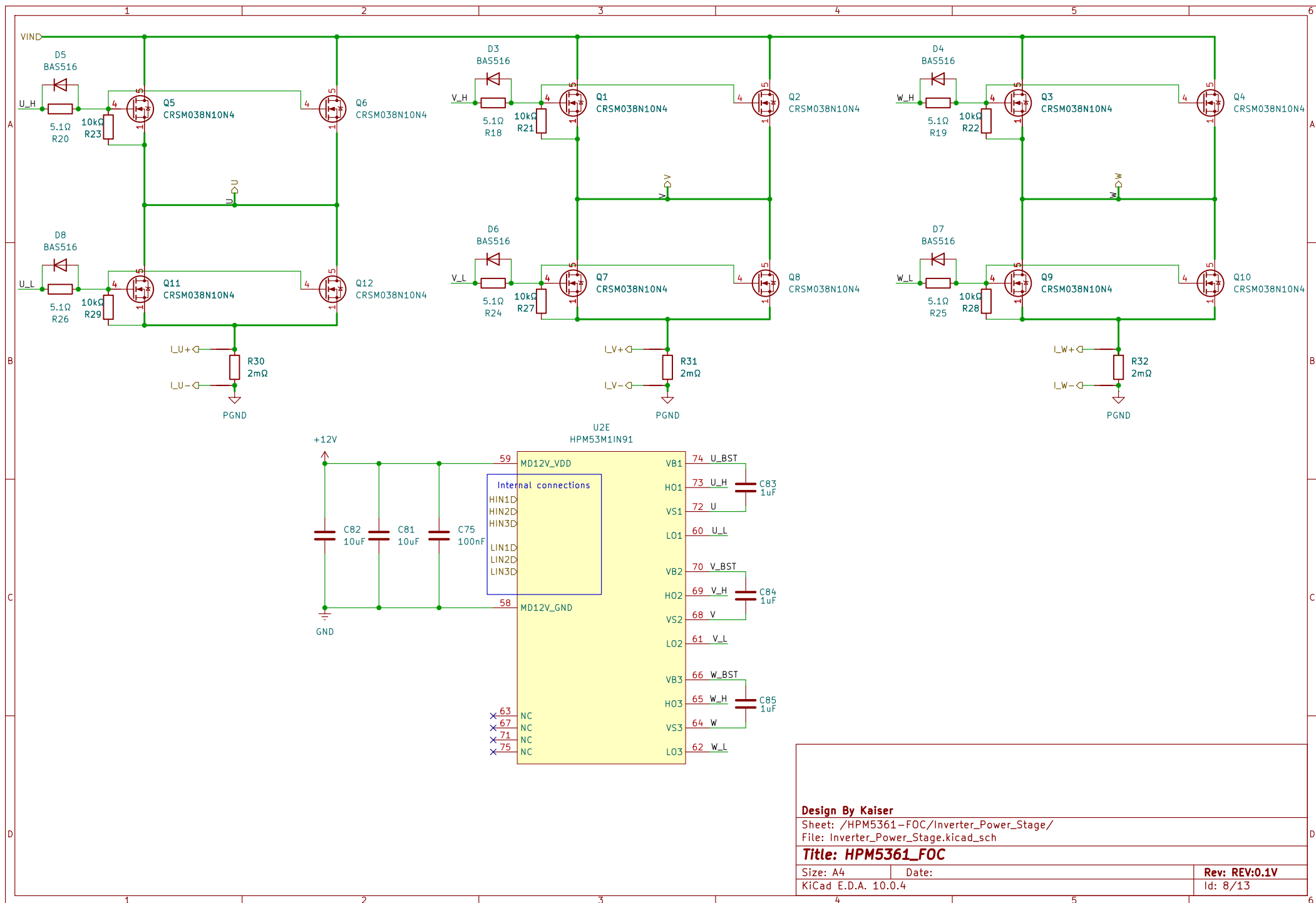
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Rev: REV:0.1V

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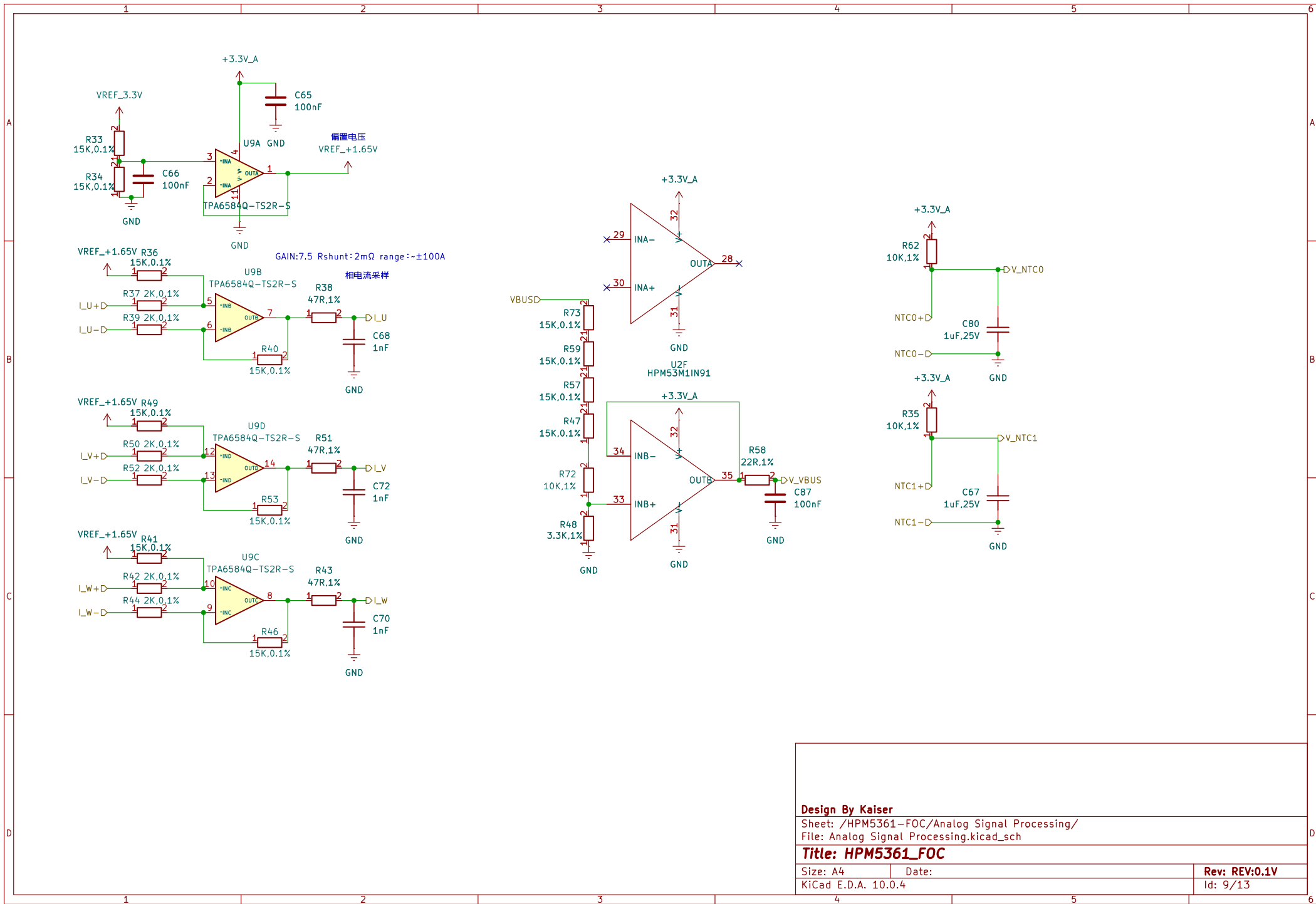
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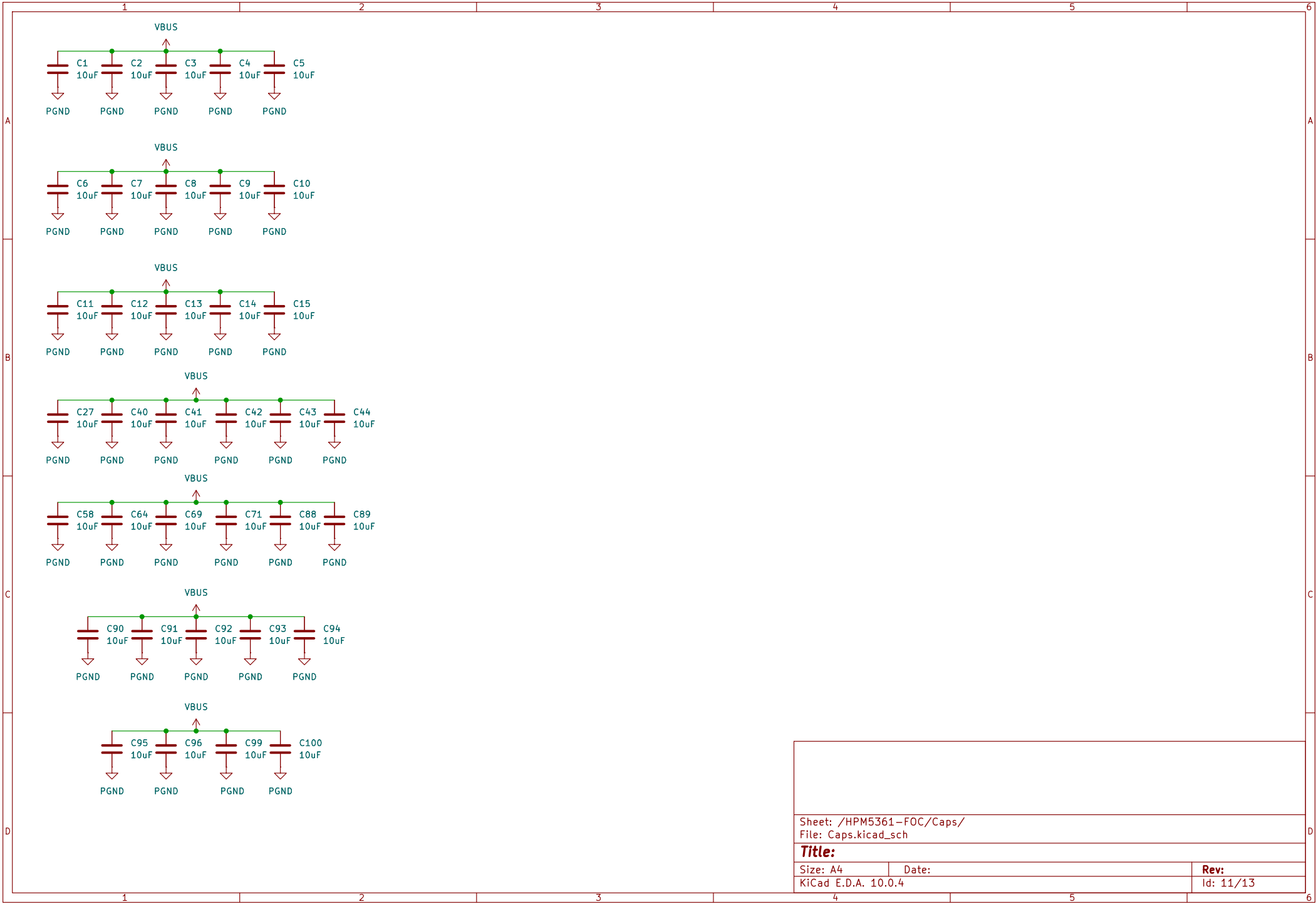
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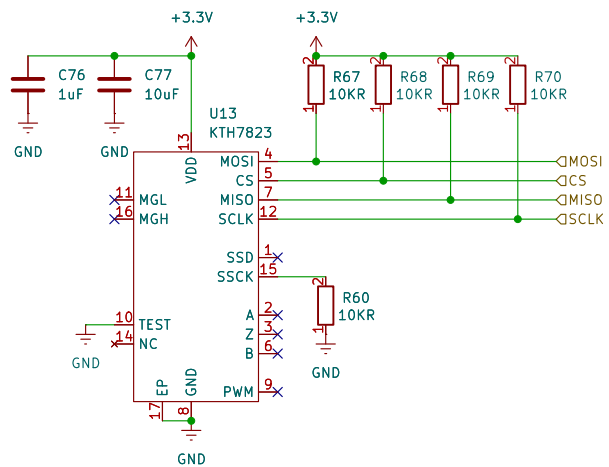
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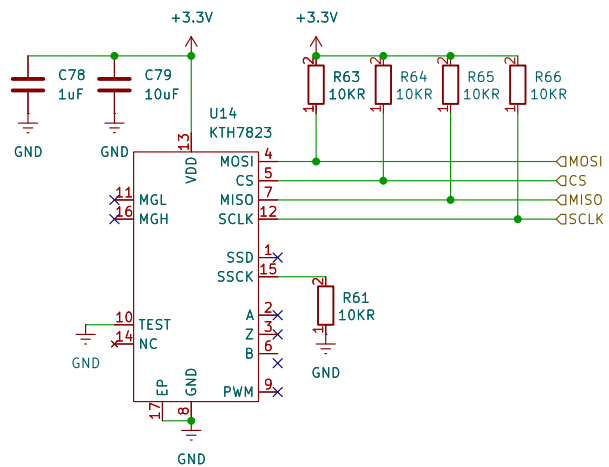
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